Technical Requirements Document (TRD)

Architecture Overview: The solution will be built natively as a mobile application compiled into an APK (Progressive Web Apps are strictly not allowed).

Tech Stack:

- **Core AI Orchestrator:** Google Antigravity (Mandatory core platform).
- **Frontend (Mobile App):** Flutter (Dart), CameraX API, Material 3 UI.
- **Backend & DB:** Firebase Firestore, Cloud Functions, Cloud Run.
- **AI Vision & Processing:** Gemini Vision API, Document OCR.
- **Notifications:** Firebase Cloud Messaging.

Agentic Workflow & Logic: The app will utilize a 5-agent architecture orchestrated by Google Antigravity:

1. **Ingestor (Document Scanner Agent):** Processes marksheet/transcript photos and extracts variables using Gemini Vision.
2. **Matcher (Scholarship Match Agent):** Compares the extracted profile against the Pakistan scholarship database using eligibility, funding, and deadline filters.
3. **Gap Detector (Analysis Agent):** Identifies missing documents and estimates the time required to obtain them.
4. **Actor (Action Execution Agent):** Triggers the 3-5 simulated actions (Drafts email, fills form, updates calendar).
5. **Monitor (Deadline Agent):** Sends push notifications based on remaining days until deadlines.

System Constraints & Compliance:

- **Budgeting:** Must operate within the $5 Google Cloud Credit limit provided for the hackathon.
- **Antigravity Traces:** The system must log and explicitly show reasoning steps, task plans, tool calls, action execution, and error recovery/fallback behavior.
- **Failure Handling:** The system must be able to demonstrate rollback or recovery if a simulated action fails (e.g., missing data)